

HTML Mastery: Beginner to Advanced

Complete Course Content — All 15 Modules

MODULE 1: Introduction to Web Development & HTML5

Lesson 1: Introduction to Web Architecture & HTML Tag Syntax

Duration: 20 mins

1.1 How the Internet Works

The internet is a global network of computers that communicate with each other. When you type a URL in your browser, here's what happens step by step:

- DNS Lookup** - Your browser asks a DNS (Domain Name System) server to convert the domain name (e.g., `google.com`) into an IP address (e.g., `142.250.64.78`).
- TCP Connection** - Your browser establishes a connection with the server using TCP/IP protocols.
- HTTP Request** - The browser sends an HTTP (HyperText Transfer Protocol) request to the server asking for the webpage.
- Server Response** - The server sends back HTML, CSS, and JavaScript files.
- Rendering** - Your browser reads and renders the files to display the webpage.

Key Protocols:

- HTTP/HTTPS** - How data is transferred between browser and server
- TCP/IP** - The underlying network communication rules
- DNS** - Converts domain names to IP addresses

1.2 Client-Server Architecture

The web runs on a **client-server model**:

Client (Browser)	Server
Sends requests	Receives and processes requests
Displays HTML/CSS/JS	Stores and serves files
Chrome, Firefox, Safari	Apache, Nginx, Node.js
Runs on user's device	Runs on remote machines

Frontend vs Backend:

- Frontend** - What users see (HTML, CSS, JavaScript)
- Backend** - Server logic and databases (PHP, Python, Node.js, SQL)

- **HTML** is a frontend technology — it defines the *structure* of a webpage
-

1.3 What is HTML?

HTML stands for **HyperText Markup Language**. It is:

- The standard language for creating webpages
- Not a programming language — it's a *markup* language
- Made up of **elements** represented by **tags**
- Read and rendered by web browsers

History of HTML:

- HTML 1.0 (1993) - Basic text and links
 - HTML 4.01 (1999) - Tables, forms, frames
 - XHTML (2000) - Stricter XML-based HTML
 - **HTML5 (2014-present)** - Semantic elements, multimedia, APIs
-

1.4 HTML Tag Syntax

Every HTML element follows this structure:

```
html

<tagname attribute="value">Content goes here</tagname>
```

Parts of an HTML element:

- `<tagname>` — **Opening tag** — starts the element
- `</tagname>` — **Closing tag** — ends the element
- `attribute="value"` — Extra information about the element
- Content — Text or other elements inside

Examples:

```
html

<h1>This is a Heading</h1>
<p class="intro">This is a paragraph.</p>
<a href="https://google.com">Click here</a>

```

Self-Closing Tags (Void Elements): Some tags have no content and close themselves:

```
html
```

```
<br />      <!-- Line break -->
<hr />    <!-- Horizontal rule -->
<img />   <!-- Image -->
<input /> <!-- Form input -->
<meta />  <!-- Metadata -->
<link />  <!-- Link to external file -->
```

1.5 The HTML DOM (Document Object Model)

The **DOM** is a tree-like representation of an HTML document. The browser builds this tree when it reads your HTML, and JavaScript can interact with it.

```
Document
├── html
│   ├── head
│   │   ├── title
│   │   └── meta
│   └── body
│       ├── h1
│       ├── p
│       └── div
│           ├── p
│           └── a
```

Every HTML element becomes a **node** in this tree. Parent-child relationships matter for CSS and JavaScript.

1.6 Setting Up VS Code

Step 1: Download VS Code Go to <https://code.visualstudio.com> and install for your OS.

Step 2: Install Extensions

- **Live Server** – Launches a local server with live reload
- **Prettier** – Auto-formats your code
- **HTML CSS Support** – Autocomplete for HTML/CSS

Step 3: Create Your First File

1. Open VS Code
2. File → New File → Save as `index.html`
3. Type `!` and press Tab — Emmet generates a full HTML boilerplate

Step 4: Launch with Live Server Right-click `index.html` → "Open with Live Server"

QUIZ — Module 1, Lesson 1 (5 Questions)

Q1. What does HTML stand for?

- A) HyperText Markup Language ☒
- B) HyperText Making Language
- C) HighText Markup Language
- D) HyperTransfer Markup Language

Q2. Which protocol converts a domain name to an IP address?

- A) HTTP
- B) TCP
- C) DNS ☒
- D) FTP

Q3. What is the correct syntax for an HTML element with an attribute?

- A) `<tagname>attribute="value"</tagname>`
- B) `<tagname attribute="value">Content</tagname>` ☒
- C) `<tagname(attribute="value")>Content</tagname>`
- D) `[tagname attribute="value"]Content[/tagname]`

Q4. Which of the following is a self-closing (void) tag?

- A) `<p>`
- B) `<div>`
- C) `<img />` ☒
- D) `<section>`

Q5. What does the DOM stand for?

- A) Document Object Model ☒
- B) Data Object Management
- C) Document Operation Markup
- D) Digital Object Model

Lesson 2: HTML Boilerplate, Meta Tags & Document Head

Duration: 25 mins

2.1 The HTML5 Boilerplate

Every HTML document starts with a standard structure called the **boilerplate**:

```
html
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>My Webpage</title>
</head>
<body>

  <h1>Hello, World!</h1>

</body>
</html>
```

Let's break down each part:

2.2 DOCTYPE Declaration

```
html

<!DOCTYPE html>
```

- Tells the browser this is an **HTML5** document
 - Must be the very **first line** — before anything else
 - Not an HTML tag — it's an instruction to the browser
 - In older HTML versions this was long and complex; HTML5 simplified it
-

2.3 The `<html>` Element

```
html

<html lang="en">
```

- The **root element** — everything else lives inside it
 - `lang="en"` declares the page language (important for accessibility and SEO)
 - Use language codes: `en` (English), `ne` (Nepali), `fr` (French), etc.
-

2.4 The `<head>` Element

The `<head>` contains **metadata** — information about the page that is NOT displayed on screen.

```
html
```

```
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Learn HTML from scratch" />
  <meta name="author" content="Pankaj Baduwal" />
  <title>HTML Mastery Course</title>
  <link rel="stylesheet" href="style.css" />
  <link rel="icon" href="favicon.ico" />
</head>
```

Key Meta Tags Explained:

Tag	Purpose
<code><meta charset="UTF-8"></code>	Supports all characters including Nepali, emojis
<code><meta name="viewport"...></code>	Makes page responsive on mobile devices
<code><meta name="description"...></code>	Search engine result description
<code><meta name="author"...></code>	Page author name
<code><title></code>	Browser tab title and SEO title

2.5 The `<body>` Element

```
html

<body>
  <!-- All visible content goes here -->
  <h1>Welcome!</h1>
  <p>This is my first webpage.</p>
</body>
```

- Everything users **see on screen** goes inside `<body>`
- Text, images, links, forms, videos — all go here
- HTML comments use `<!-- comment -->` syntax and are NOT shown on screen

2.6 HTML Comments

```
html

<!-- This is a comment and will not appear on the page -->
<p>This paragraph will appear.</p>
<!-- TODO: Add navigation here -->
```

Comments are useful for:

- Explaining your code
 - Temporarily hiding content during development
 - Leaving notes for yourself or team members
-

2.7 Linking External Files

```
html

<!-- Link CSS stylesheet -->
<link rel="stylesheet" href="style.css" />

<!-- Link JavaScript file (placed at end of body) -->
<script src="script.js"></script>

<!-- Link a favicon (browser tab icon) -->
<link rel="icon" type="image/png" href="favicon.png" />
```

QUIZ — Module 1, Lesson 2 (5 Questions)

Q1. Where must the `<!DOCTYPE html>` declaration appear?

- A) Inside the `<head>` tag
- B) After the `<html>` tag
- C) As the very first line of the document ☒
- D) Inside the `<body>` tag

Q2. Which meta tag makes a webpage mobile-friendly?

- A) `<meta charset="UTF-8">`
- B) `<meta name="viewport" content="width=device-width, initial-scale=1.0">` ☒
- C) `<meta name="description">`
- D) `<meta name="author">`

Q3. What is the purpose of the `<head>` section in HTML?

- A) To display the page header on screen
- B) To contain metadata not shown on the page ☒
- C) To create the navigation bar
- D) To define the page body

Q4. How do you write an HTML comment?

- A) `// This is a comment`
- B) `/* This is a comment */`
- C) `<!-- This is a comment -->` ☒

- D) `# This is a comment`

Q5. Which tag is used to set the browser tab title?

- A) `<meta name="title">`
- B) `<header>`
- C) `<h1>`
- D) `<title>` 

MODULE 2: Text Elements & Formatting

Lesson 1: Headings, Paragraphs & Text Styling

Duration: 20 mins

2.1 Headings — `<h1>` to `<h6>`

HTML provides six levels of headings:

```
html

<h1>Heading 1 – Main Title (Largest)</h1>
<h2>Heading 2 – Section Title</h2>
<h3>Heading 3 – Sub-section</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6 – Smallest Heading</h6>
```

Rules for Headings:

- Only **one** `<h1>` per page (important for SEO)
- Use headings in **hierarchical order** — don't skip levels
- Headings communicate structure, not just size
- Screen readers use headings to navigate pages

2.2 Paragraphs — `<p>`

```
html

<p>This is a paragraph. It contains a block of text.</p>
<p>Each paragraph starts on a new line with automatic spacing.</p>
```

- Browsers automatically add space above and below `<p>` tags
- Whitespace inside HTML collapses — multiple spaces become one

2.3 Line Break & Horizontal Rule

```
html
```



```
<p>This is line one.<br />This is line two in the same paragraph.</p>
```

```
<hr />  <!-- Creates a horizontal dividing line -->
```

- `<br />` — Forces a line break within text (use sparingly)
- `<hr />` — Creates a thematic break/divider between sections

2.4 Text Formatting Tags

html

```
<!-- Bold (semantic importance) -->
```

```
<strong>This text is important</strong>
```

```
<!-- Bold (visual only) -->
```

```
<b>This text is bold</b>
```

```
<!-- Italic (semantic emphasis) -->
```

```
<em>This text is emphasized</em>
```

```
<!-- Italic (visual only) -->
```

```
<i>This text is italic</i>
```

```
<!-- Underline -->
```

```
<u>This text is underlined</u>
```

```
<!-- Strikethrough -->
```

```
<s>This text is crossed out</s>
```

```
<del>This was deleted</del>
```

```
<!-- Superscript and Subscript -->
```

```
<p>H<sub>2</sub>O is water. E=mc<sup>2</sup> is Einstein's equation.</p>
```

```
<!-- Highlighted text -->
```

```
<mark>This text is highlighted</mark>
```

```
<!-- Small text -->
```

```
<small>Copyright 2024</small>
```

```
<!-- Inline code -->
```

```
<code>console.log("hello")</code>
```

Semantic vs Presentational:

- `<strong>` and `<em>` carry **meaning** (used by screen readers)
- `<b>` and `<i>` are **visual only** — prefer `<strong>` and `<em>`

2.5 Quotations

```
html

<!-- Block quote (long quotation from another source) -->
<blockquote cite="https://example.com">
  <p>"The best way to predict the future is to create it."</p>
</blockquote>

<!-- Inline quote -->
<p>He said <q>HTML is the backbone of the web.</q></p>

<!-- Citation / reference -->
<p><cite>HTML5 Specification</cite> was published by W3C.</p>

<!-- Abbreviation with tooltip -->
<p><abbr title="HyperText Markup Language">HTML</abbr> is easy to learn.</p>
```

2.6 Preformatted Text

```
html

<pre>
  This text
    preserves    all spacing
  and line breaks exactly.
</pre>
```

- `<pre>` displays text in a **fixed-width font**
- Preserves all **whitespace and line breaks**
- Useful for code samples and ASCII art

2.7 `<div>` and `<span>` — Generic Containers

```
html

<!-- Block-level container (takes full width) -->
<div class="card">
  <h2>Card Title</h2>
  <p>Card content here.</p>
</div>

<!-- Inline container (stays within text flow) -->
<p>My favorite color is <span style="color:blue">blue</span>.</p>
```

- `<div>` — **block-level** container, starts on new line

- `<span>` — **inline** container, stays within the text
 - Neither has semantic meaning — they're just for grouping/styling
-

QUIZ — Module 2 (5 Questions)

Q1. How many `<h1>` tags should appear on a single webpage?

- A) As many as needed
- B) Maximum 3
- C) Only one 
- D) Only two

Q2. Which tag is used for semantic importance (not just visual bold)?

- A) `<b>`
- B) `<strong>` 
- C) `<bold>`
- D) `<heavy>`

Q3. What does the `<pre>` tag do?

- A) Adds a preview section
- B) Makes text bold
- C) Preserves whitespace and line breaks as written 
- D) Creates a preloader

Q4. What is the difference between `<div>` and `<span>`?

- A) `<div>` is for images, `<span>` is for text
- B) `<div>` is block-level, `<span>` is inline 
- C) `<div>` is inline, `<span>` is block-level
- D) They are identical

Q5. Which tag represents text that has been deleted (strikethrough with semantic meaning)?

- A) `<s>`
 - B) `<strike>`
 - C) `<del>` 
 - D) `<remove>`
-

MODULE 3: Links & Navigation Architecture

Lesson 1: Hyperlinks & Navigation Anchors

Duration: 25 mins

3.1 The Anchor Tag — `<a>`

The `<a>` (anchor) tag creates hyperlinks:

```
html

<a href="https://www.google.com">Visit Google</a>
```

Parts:

- `href` — "HyperText REference" — the destination URL
 - Link text — what the user clicks on
-

3.2 Absolute vs Relative Paths

Absolute URLs — full address including protocol:

```
html

<a href="https://www.example.com/about.html">About Us</a>
<a href="https://www.youtube.com">YouTube</a>
```

Relative URLs — relative to the current file's location:

```
html

<!-- Same folder -->
<a href="about.html">About</a>

<!-- Subfolder -->
<a href="pages/contact.html">Contact</a>

<!-- Go up one folder -->
<a href="../index.html">Home</a>

<!-- Root of site -->
<a href="/index.html">Home</a>
```

3.3 The `target` Attribute

```
html
```

```
<!-- Open in same tab (default) -->
<a href="about.html" target="_self">About</a>

<!-- Open in new tab -->
<a href="https://google.com" target="_blank" rel="noopener noreferrer">Google</a>

<!-- Open in parent frame -->
<a href="page.html" target="_parent">Parent</a>
```

Security Note: Always add `rel="noopener noreferrer"` when using `target="_blank"` to prevent security vulnerabilities.

3.4 Email and Phone Links

```
html

<!-- Email link (opens default mail app) -->
<a href="mailto:pankaj@example.com">Send Email</a>

<!-- Email with subject and body pre-filled -->
<a href="mailto:info@site.com?subject=Hello&body=I%20want%20to%20enroll">Contact Us</a>

<!-- Phone link (clickable on mobile) -->
<a href="tel:+9779800000000">Call Us: +977-980-000-0000</a>
```

3.5 Bookmark Links (Jump to Section)

```
html

<!-- Create a bookmark target using id -->
<h2 id="courses">Our Courses</h2>
<h2 id="contact">Contact Us</h2>

<!-- Link to the bookmark -->
<a href="#courses">Jump to Courses</a>
<a href="#contact">Jump to Contact</a>

<!-- Link to bookmark on another page -->
<a href="about.html#team">Meet the Team</a>
```

3.6 Navigation Menu Structure

```
html
```

```
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="courses.html">Courses</a></li>
    <li><a href="contact.html">Contact</a></li>
  </ul>
</nav>
```

3.7 Download Links

```
html

<!-- Link to download a file -->
<a href="files/resume.pdf" download>Download My Resume</a>

<!-- Download with a custom filename -->
<a href="files/resume.pdf" download="Pankaj-Resume.pdf">Download Resume</a>
```

QUIZ — Module 3 (5 Questions)

Q1. What does `href` stand for in an anchor tag?

- A) HyperText Reference File
- B) HyperText REference ☒
- C) HyperText Render Format
- D) HyperText Reading Format

Q2. Which attribute opens a link in a new browser tab?

- A) `target="_new"`
- B) `target="_tab"`
- C) `target="_blank"` ☒
- D) `open="new"`

Q3. What type of URL is `../pages/about.html`?

- A) Absolute URL
- B) Relative URL ☒
- C) Bookmark URL
- D) Protocol URL

Q4. How do you create a clickable phone link?

- A) `<a href="phone:+977...">Call</a>`
- B) `<a href="call:+977...">Call</a>`

- C) `<a href="tel:+977...">Call</a>` ✓
- D) `<a href="mobile:+977...">Call</a>`

Q5. What security attribute should be added when using `target="_blank"`?

- A) `rel="nofollow"`
- B) `rel="noopener noreferrer"` ✓
- C) `rel="safe"`
- D) `secure="true"`

MODULE 4: Images & Multimedia Integration

Lesson 1: Images, Audio, Video & iFrames

Duration: 30 mins

4.1 The `<img>` Tag

```
html

```

Key Attributes:

Attribute	Purpose
<code>src</code>	Source — path to the image file
<code>alt</code>	Alternative text (accessibility & SEO)
<code>width</code> / <code>height</code>	Dimensions in pixels
<code>loading="lazy"</code>	Defers loading until image is in viewport
<code>title</code>	Tooltip text on hover

Image Formats:

- **JPEG/JPG** — Photos, complex images (lossy compression)
 - **PNG** — Images needing transparency
 - **GIF** — Simple animations
 - **SVG** — Vector graphics (infinitely scalable)
 - **WebP** — Modern format, best compression + quality
-

4.2 The `alt` Attribute — Critical for Accessibility

```
html
```

```
<!-- Good alt text: descriptive -->


<!-- Decorative image: empty alt (screen reader skips it) -->


<!-- Bad: missing alt (accessibility failure) -->

```

4.3 Figure and Figcaption

```
html

<figure>
  
  <figcaption>The official HTML5 logo, adopted in 2011</figcaption>
</figure>
```

`<figure>` groups media with its caption. `<figcaption>` provides a visible caption.

4.4 SVG Inline

```
html

<!-- Simple inline SVG circle -->
<svg width="100" height="100" xmlns="http://www.w3.org/2000/svg">
  <circle cx="50" cy="50" r="40" fill="blue" />
</svg>

<!-- SVG as image file -->

```

SVG = Scalable Vector Graphics — perfect for logos, icons, illustrations.

4.5 HTML5 Audio — `<audio>`

```
html

<audio controls>
  <source src="music.mp3" type="audio/mpeg" />
  <source src="music.ogg" type="audio/ogg" />
  Your browser does not support the audio element.
</audio>
```

Attributes:

- `controls` — Shows play/pause/volume controls
 - `autoplay` — Starts playing automatically (use sparingly)
 - `loop` — Loops the audio
 - `muted` — Starts muted
-

4.6 HTML5 Video — `<video>`

html

```
<video width="640" height="360" controls poster="thumbnail.jpg">
  <source src="intro.mp4" type="video/mp4" />
  <source src="intro.webm" type="video/webm" />
  Your browser does not support the video element.
</video>
```

Attributes:

- `controls` — Shows video controls
 - `poster` — Thumbnail image shown before play
 - `autoplay muted` — Auto-plays silently (browsers allow this combination)
 - `loop` — Loops the video
 - `preload="none|metadata|auto"` — Controls how much to preload
-

4.7 iFrames — Embedding External Content

html

```
<!-- Embed a YouTube video -->
<iframe
  width="560"
  height="315"
  src="https://www.youtube.com/embed/VIDEO_ID"
  title="YouTube video player"
  allowfullscreen
  frameborder="0">
</iframe>

<!-- Embed Google Maps -->
<iframe
  src="https://www.google.com/maps/embed?pb=..."
  width="600"
  height="450"
  allowfullscreen
  loading="lazy">
</iframe>
```

QUIZ — Module 4 (5 Questions)

Q1. Which attribute provides alternative text for an image?

- A) `title`
- B) `description`
- C) `alt` ✓
- D) `caption`

Q2. Which image format supports transparency?

- A) JPEG
- B) PNG ✓
- C) GIF
- D) BMP

Q3. What does the `controls` attribute do in `<video>` and `<audio>` tags?

- A) Sets volume automatically
- B) Shows the browser's built-in media player controls ✓
- C) Controls the frame rate
- D) Enables autoplay

Q4. Which tag is used to embed external content like YouTube videos?

- A) `<embed>`
- B) `<object>`
- C) `<frame>`
- D) `<iframe>` ✓

Q5. What does the `poster` attribute in `<video>` do?

- A) Adds a watermark
- B) Shows a thumbnail image before the video plays ✓
- C) Sets the video title
- D) Adds captions

MODULE 5: Lists (Unordered, Ordered & Description Lists)

Lesson 1: List Types & Nested Lists

Duration: 20 mins

5.1 Unordered Lists — `<ul>`

Used for items with **no particular order**:

```
html
```

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

Default display: bullet points (●)

```
html

<!-- Change bullet style with CSS -->
<ul style="list-style-type: square;">
  <li>Square bullets</li>
</ul>

<ul style="list-style-type: none;">
  <li>No bullets (common for navigation)</li>
</ul>
```

5.2 Ordered Lists — `<ol>`

Used for items in a **specific sequence**:

```
html

<ol>
  <li>Learn HTML</li>
  <li>Learn CSS</li>
  <li>Learn JavaScript</li>
  <li>Build projects</li>
</ol>
```

Attributes for `<ol>`:

```
html
```

```
<!-- Start from number 5 -->
<ol start="5">
  <li>Fifth item</li>
  <li>Sixth item</li>
</ol>

<!-- Reverse order -->
<ol reversed>
  <li>Last step</li>
  <li>Second-to-last</li>
</ol>

<!-- Use letters -->
<ol type="A">
  <li>Option A</li>
  <li>Option B</li>
</ol>

<!-- Use Roman numerals -->
<ol type="I">
  <li>Introduction</li>
  <li>Methods</li>
</ol>
```

5.3 Description Lists — `<dl>`

Used for **term-definition pairs** (like a glossary):

```
html

<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language – used to structure webpages</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets – used to style webpages</dd>

  <dt>JavaScript</dt>
  <dd>A programming language that makes webpages interactive</dd>
</dl>
```

- `<dl>` — Description List container
 - `<dt>` — Description Term (the word/term)
 - `<dd>` — Description Detail (the definition/explanation)
-

5.4 Nested Lists

Lists can be placed inside list items:

```
html

<ul>
  <li>Frontend Technologies
    <ul>
      <li>HTML</li>
      <li>CSS
        <ul>
          <li>Flexbox</li>
          <li>Grid</li>
        </ul>
      </li>
      <li>JavaScript</li>
    </ul>
  </li>
  <li>Backend Technologies
    <ol>
      <li>Node.js</li>
      <li>Python</li>
      <li>PHP</li>
    </ol>
  </li>
</ul>
```

5.5 Navigation Using Lists

Lists are commonly used for navigation menus:

```
html

<nav>
  <ul>
    <li><a href="#home">Home</a></li>
    <li><a href="#about">About</a>
      <ul>
        <li><a href="#team">Team</a></li>
        <li><a href="#history">History</a></li>
      </ul>
    </li>
    <li><a href="#contact">Contact</a></li>
  </ul>
</nav>
```

QUIZ — Module 5 (5 Questions)

Q1. Which tag creates an unordered (bulleted) list?

- A) `<ol>`
- B) `<list>`
- C) `<ul>` ✓
- D) `<li>`

Q2. How do you make an ordered list start from number 5?

- A) `<ol begin="5">`
- B) `<ol start="5">` ✓
- C) `<ol from="5">`
- D) `<ol number="5">`

Q3. What are the three tags used in a Description List?

- A) `<dl>`, `<di>`, `<dd>`
- B) `<dl>`, `<dt>`, `<dd>` ✓
- C) `<list>`, `<term>`, `<def>`
- D) `<dl>`, `<dt>`, `<def>`

Q4. Which `<ol>` type attribute value shows Roman numerals?

- A) `type="R"`
- B) `type="roman"`
- C) `type="I"` ✓
- D) `type="num"`

Q5. Can a `<ul>` list be placed inside an `<ol>` list item?

- A) No, lists cannot be nested
- B) Yes, but only two levels deep
- C) Yes, lists can be nested inside each other ✓
- D) Only `<ol>` can be nested inside `<ul>`

MODULE 6: Tables & Complex Tabular Data

Lesson 1: Building Data Tables with Rowspan & Colspan

Duration: 25 mins

6.1 Basic Table Structure

html

```
<table border="1">
  <thead>
    <tr>
      <th>Name</th>
      <th>Course</th>
      <th>Score</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Pankaj</td>
      <td>HTML Mastery</td>
      <td>95</td>
    </tr>
    <tr>
      <td>Anita</td>
      <td>HTML Mastery</td>
      <td>88</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan="2">Average Score</td>
      <td>91.5</td>
    </tr>
  </tfoot>
</table>
```

Table Tags:

Tag	Purpose
<table>	Container for the entire table
<thead>	Table header section
<tbody>	Table body section
<tfoot>	Table footer section
<tr>	Table row
<th>	Table header cell (bold, centered by default)
<td>	Table data cell

6.2 The `colspan` Attribute

Spans a cell across **multiple columns**:

html

```
<table border="1">
  <tr>
    <th colspan="3">Student Report Card</th>
  </tr>
  <tr>
    <th>Subject</th>
    <th>Marks</th>
    <th>Grade</th>
  </tr>
  <tr>
    <td>HTML</td>
    <td>90</td>
    <td>A</td>
  </tr>
</table>
```

6.3 The `rowspan` Attribute

Spans a cell across **multiple rows**:

html

```
<table border="1">
  <tr>
    <th>Day</th>
    <th>Time</th>
    <th>Subject</th>
  </tr>
  <tr>
    <td rowspan="2">Monday</td>
    <td>9:00 AM</td>
    <td>HTML</td>
  </tr>
  <tr>
    <td>11:00 AM</td>
    <td>CSS</td>
  </tr>
</table>
```


6.4 Table Caption

```
html

<table>
  <caption>Monthly Sales Report – 2024</caption>
  <thead>...</thead>
  <tbody>...</tbody>
</table>
```

6.5 Table Accessibility

```
html

<table>
  <caption>Course Schedule</caption>
  <thead>
    <tr>
      <th scope="col">Course</th>
      <th scope="col">Duration</th>
      <th scope="col">Level</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <th scope="row">HTML Mastery</th>
      <td>24 hours</td>
      <td>Beginner</td>
    </tr>
  </tbody>
</table>
```

`scope="col"` and `scope="row"` help screen readers understand table structure.

QUIZ — Module 6 (5 Questions)

Q1. Which tag defines a table header cell?

- A) `<td>`
- B) `<th>` ✓
- C) `<thead>`
- D) `<header>`

Q2. What does `colspan="3"` do?

- A) Adds 3 rows to a cell
- B) Makes a cell span across 3 columns ✓

- C) Limits a cell to 3 characters
- D) Creates 3 new cells

Q3. Which section of a table should contain totals or summary rows?

- A) `<thead>`
- B) `<tbody>`
- C) `<tfoot>` ✓
- D) `<tsummary>`

Q4. What tag adds a title/description to a table?

- A) `<title>`
- B) `<label>`
- C) `<caption>` ✓
- D) `<summary>`

Q5. What is `rowspan="2"` used for?

- A) Making text span 2 lines
- B) Making a cell span across 2 rows ✓
- C) Adding 2 new rows
- D) Merging 2 tables

MODULE 7: Interactive HTML5 Forms & Controls

Lesson 1: User Input Forms & Native Validation

Duration: 30 mins

7.1 The Form Element

```
html

<form action="/submit" method="POST">
  <!-- form controls go here -->
</form>
```

Key Attributes:

- `action` — URL where form data is sent
 - `method` — `GET` (data in URL) or `POST` (data in request body)
-

7.2 Text Input Fields

```
html
```

```
<!-- Text input -->
<label for="name">Full Name:</label>
<input type="text" id="name" name="name" placeholder="Enter your name" required />

<!-- Email input (validates email format) -->
<label for="email">Email:</label>
<input type="email" id="email" name="email" required />

<!-- Password input (hides characters) -->
<label for="password">Password:</label>
<input type="password" id="password" name="password" minlength="8" required />

<!-- Number input -->
<input type="number" name="age" min="1" max="120" />

<!-- Phone number -->
<input type="tel" name="phone" pattern="[0-9]{10}" />

<!-- URL input -->
<input type="url" name="website" placeholder="https://example.com" />

<!-- Search box -->
<input type="search" name="q" placeholder="Search..." />
```

7.3 Labels — Critical for Accessibility

```
html

<!-- Method 1: Using 'for' and 'id' (recommended) -->
<label for="username">Username:</label>
<input type="text" id="username" name="username" />

<!-- Method 2: Wrapping input inside label -->
<label>
  Email:
  <input type="email" name="email" />
</label>
```

Always pair `<label>` with input fields — this is required for accessibility.

7.4 Textarea — Multi-line Text

```
html
```

```
<label for="message">Message:</label>
<textarea id="message" name="message" rows="5" cols="40" placeholder="Write you
```

7.5 Radio Buttons

```
html

<fieldset>
  <legend>Select your level:</legend>

  <input type="radio" id="beginner" name="level" value="beginner" checked />
  <label for="beginner">Beginner</label>

  <input type="radio" id="intermediate" name="level" value="intermediate" />
  <label for="intermediate">Intermediate</label>

  <input type="radio" id="advanced" name="level" value="advanced" />
  <label for="advanced">Advanced</label>
</fieldset>
```

Radio buttons with the **same** `name` form a group — only one can be selected.

7.6 Checkboxes

```
html

<fieldset>
  <legend>Select your interests:</legend>

  <input type="checkbox" id="html" name="interest" value="html" checked />
  <label for="html">HTML</label>

  <input type="checkbox" id="css" name="interest" value="css" />
  <label for="css">CSS</label>

  <input type="checkbox" id="js" name="interest" value="js" />
  <label for="js">JavaScript</label>
</fieldset>
```

7.7 Select Dropdowns

```
html
```

```
<label for="country">Country:</label>
<select id="country" name="country">
  <option value="">-- Select Country --</option>
  <optgroup label="South Asia">
    <option value="np">Nepal</option>
    <option value="in">India</option>
    <option value="bd">Bangladesh</option>
  </optgroup>
  <optgroup label="Europe">
    <option value="uk">United Kingdom</option>
    <option value="de">Germany</option>
  </optgroup>
</select>
```

7.8 Other Input Types

```
html

<!-- Date picker -->
<input type="date" name="birthday" />

<!-- Color picker -->
<input type="color" name="theme" value="#ff0000" />

<!-- Range slider -->
<input type="range" name="volume" min="0" max="100" value="50" />

<!-- File upload -->
<input type="file" name="document" accept=".pdf,.doc" multiple />

<!-- Hidden field (submitted but not shown) -->
<input type="hidden" name="user_id" value="12345" />
```

7.9 Form Buttons

```
html
```

```
<!-- Submit button -->
<button type="submit">Submit Form</button>

<!-- Reset button (clears form) -->
<button type="reset">Clear</button>

<!-- Regular button (for JavaScript) -->
<button type="button" onclick="doSomething()">Click Me</button>

<!-- Input submit (older style) -->
<input type="submit" value="Send" />
```

7.10 Native HTML5 Validation

```
html

<form>
  <!-- Required field -->
  <input type="text" required />

  <!-- Minimum/maximum length -->
  <input type="text" minlength="3" maxlength="50" />

  <!-- Pattern matching (regex) -->
  <input type="text" pattern="[A-Za-z]{3,}" title="Only letters, minimum 3 char" />

  <!-- Min/max for numbers -->
  <input type="number" min="18" max="99" />

  <button type="submit">Submit</button>
</form>
```

QUIZ — Module 7 (5 Questions)

Q1. What is the correct way to link a `<label>` to an `<input>` field?

- A) Use the `name` attribute on both
- B) Use `for` on the label matching the `id` on the input ☒
- C) Place the input before the label
- D) Use the `link` attribute

Q2. How do you make radio buttons in a group exclusive (only one selectable)?

- A) Give them the same `id`
- B) Give them the same `name` ☒

- C) Wrap them in `<radiogroup>`
- D) Add `exclusive="true"`

Q3. Which input type automatically validates email format?

- A) `type="text"`
- B) `type="mail"`
- C) `type="email"` ✓
- D) `type="address"`

Q4. What HTML attribute makes a form field mandatory?

- A) `mandatory`
- B) `validate`
- C) `compulsory`
- D) `required` ✓

Q5. What does `<input type="hidden">` do?

- A) Makes input invisible but still submitted ✓
- B) Encrypts the field value
- C) Prevents the field from being submitted
- D) Creates a password field

MODULE 8: Semantic HTML5 Architecture

Lesson 1: Semantic Tags & Page Layout

Duration: 25 mins

8.1 What is Semantic HTML?

Semantic HTML means using tags that convey meaning about the content they contain — not just how it looks.

Non-semantic (bad):

```
html
<div id="header">...</div>
<div id="nav">...</div>
<div class="main-content">...</div>
<div id="footer">...</div>
```

Semantic (good):

```
html
```

```
<header>...</header>
<nav>...</nav>
<main>...</main>
<footer>...</footer>
```

Why Semantic HTML Matters:

- **Accessibility** — Screen readers understand page structure
 - **SEO** — Search engines better understand content
 - **Maintainability** — Code is easier to read and maintain
 - **Standards Compliance** — Follows W3C best practices
-

8.2 Page-Level Semantic Elements

html


```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Semantic Page</title>
</head>
<body>

  <header>
    <h1>My Website</h1>
    <nav>
      <ul>
        <li><a href="/">Home</a></li>
        <li><a href="/about">About</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <article>
      <h2>Blog Post Title</h2>
      <p>Published on <time datetime="2024-01-15">January 15, 2024</time></p>
      <p>Article content here...</p>
    </article>

    <aside>
      <h3>Related Articles</h3>
      <ul>
        <li><a href="#">HTML Tips</a></li>
      </ul>
    </aside>
  </main>

  <footer>
    <p>&copy; 2024 My Website. All rights reserved.</p>
  </footer>

</body>
</html>
```

8.3 Semantic Elements Reference

Element	Use For
<code><header></code>	Site header or section header (logo, nav)
<code><nav></code>	Navigation links
<code><main></code>	Primary content of the page (only one per page)
<code><article></code>	Self-contained content (blog post, news article)
<code><section></code>	Thematic grouping within a page
<code><aside></code>	Sidebar, related content, ads
<code><footer></code>	Footer of page or section
<code><figure></code>	Media (image, chart) with optional caption
<code><figcaption></code>	Caption for a figure
<code><time></code>	Date and/or time value
<code><mark></code>	Highlighted text
<code><details></code>	Expandable content section
<code><summary></code>	Visible title for a <code><details></code> block
<code><address></code>	Contact information

8.4 `<article>` vs `<section>`

`<article>` — Self-contained, could stand alone (blog post, review, comment):

```
html

<article>
  <h2>How to Learn HTML in 30 Days</h2>
  <p>Learning HTML starts with understanding structure...</p>
</article>
```

`<section>` — Thematic grouping within a page (chapter, tab):

```
html
```

```
<section id="features">
  <h2>Course Features</h2>
  <p>This course includes...</p>
</section>
```

8.5 `<details>` and `<summary>` — Native Accordion

```
html

<details>
  <summary>What is HTML?</summary>
  <p>HTML stands for HyperText Markup Language. It is the standard language for
</details>

<details>
  <summary>Is this course free?</summary>
  <p>Yes! This course is free for all PiyushDhara students.</p>
</details>
```

No JavaScript needed — this creates a collapsible FAQ section.

8.6 `<address>` — Contact Information

```
html

<address>
  <strong>Pankaj Baduwal</strong><br />
  Lead Educator<br />
  <a href="mailto:pankaj@example.com">pankaj@example.com</a><br />
  Kathmandu, Nepal
</address>
```

QUIZ — Module 8 (5 Questions)

Q1. Which element should contain the primary content of a webpage?

- A) `<content>`
- B) `<body>`
- C) `<main>` ✓
- D) `<section>`

Q2. What is the difference between `<article>` and `<section>`?

- A) They are identical
- B) `<article>` is self-contained content; `<section>` is a thematic grouping ✓

- C) `<section>` is self-contained; `<article>` is a grouping
- D) `<article>` is only for news

Q3. Which element creates a native expandable/collapsible section without JavaScript?

- A) `<expand>`
- B) `<toggle>`
- C) `<details>` 
- D) `<collapse>`

Q4. How many `<main>` elements should appear on a single page?

- A) As many as needed
- B) Only one 
- C) Two maximum
- D) One per section

Q5. Which tag is used for navigation menus?

- A) `<navigation>`
- B) `<menu>`
- C) `<nav>` 
- D) `<links>`

MODULE 9: Web Accessibility (A11Y) & ARIA Standards

Lesson 1: Accessible Web Standards & ARIA Roles

Duration: 25 mins

9.1 What is Web Accessibility?

Web accessibility (A11Y) means ensuring websites can be used by everyone — including people with:

- Visual impairments (use screen readers like NVDA, VoiceOver)
- Motor disabilities (use keyboard only, voice control)
- Hearing impairments
- Cognitive disabilities

WCAG 2.1 (Web Content Accessibility Guidelines) defines four principles:

- **Perceivable** — Content can be seen/heard by all
 - **Operable** — All functions work via keyboard
 - **Understandable** — Content is clear and consistent
 - **Robust** — Works with current and future assistive tech
-

9.2 Text Alternatives (Alt Text)

```
html

<!-- Informative image: describe what it shows -->



<!-- Complex image: link to longer description -->

<button>First</button>
<button>Second</button>
<button>Third</button>

<!-- Custom tab order (use sparingly) -->
<button tabindex="2">Second</button>
<button tabindex="1">First</button>

<!-- Remove from tab order -->
<div tabindex="-1" id="modal">Modal content</div>

<!-- Make a non-interactive element focusable -->
<div tabindex="0" role="button" onclick="activate()">Custom Button</div>
```

9.4 ARIA — Accessible Rich Internet Applications

ARIA attributes add accessibility information that HTML alone can't provide.

ARIA Roles:

```
html
```

```
<!-- Navigation landmark -->
<nav role="navigation" aria-label="Main navigation">...</nav>

<!-- Main content landmark -->
<main role="main">...</main>

<!-- Banner landmark -->
<header role="banner">...</header>

<!-- Content info landmark -->
<footer role="contentinfo">...</footer>

<!-- Custom button -->
<div role="button" tabindex="0" aria-pressed="false">Toggle</div>

<!-- Alert (announced immediately by screen readers) -->
<div role="alert">Form submitted successfully!</div>

<!-- Dialog (modal) -->
<div role="dialog" aria-labelledby="dialog-title" aria-modal="true">
  <h2 id="dialog-title">Confirm Action</h2>
</div>
```

9.5 ARIA States and Properties

html

```

<!-- aria-label: provides a label when visible text is not enough -->
<button aria-label="Close dialog">X</button>

<!-- aria-labelledby: points to another element as the label -->
<section aria-labelledby="section-title">
  <h2 id="section-title">Our Courses</h2>
</section>

<!-- aria-describedby: additional description -->
<input type="password" aria-describedby="pwd-hint" />
<p id="pwd-hint">Password must be at least 8 characters.</p>

<!-- aria-expanded: shows if something is open/closed -->
<button aria-expanded="false" aria-controls="dropdown">Menu</button>
<ul id="dropdown" hidden>...</ul>

<!-- aria-hidden: hides from screen readers -->
<span aria-hidden="true">🎉</span> Congratulations!

<!-- aria-required: marks required fields -->
<input type="text" aria-required="true" />

<!-- aria-live: announces dynamic changes -->
<div aria-live="polite" aria-atomic="true">
  Loading results...
</div>

```

9.6 Skip Navigation Link

html

```

<!-- First element in body – lets keyboard users skip to content -->
<a href="#main-content" class="skip-link">Skip to main content</a>

<header>...</header>

<main id="main-content">
  <h1>Welcome</h1>
</main>

```

CSS

```
.skip-link {
  position: absolute;
  top: -40px;
  left: 0;
  background: #000;
  color: #fff;
  padding: 8px;
  z-index: 100;
}
.skip-link:focus {
  top: 0;
}
```

9.7 Accessible Forms

```
html

<form>
  <!-- Always use labels -->
  <label for="email">Email Address <span aria-hidden="true">*</span></label>
  <input type="email" id="email" name="email" required aria-required="true" />

  <!-- Error messages linked to field -->
  <input type="text" id="name" aria-describedby="name-error" aria-invalid="true"
  <span id="name-error" role="alert">Please enter your full name.</span>

  <!-- Fieldset groups related inputs -->
  <fieldset>
    <legend>Payment Method</legend>
    <input type="radio" id="card" name="payment" value="card" />
    <label for="card">Credit Card</label>
  </fieldset>
</form>
```

QUIZ — Module 9 (5 Questions)

Q1. What does A11Y stand for?

- A) Accessibility ☒
- B) Algorithm Year 11
- C) Audio Layer Year
- D) Automated Layout Year

Q2. What ARIA role announces updates to screen readers immediately?

- A) `role="live"`

- B) `role="alert"` ✓
- C) `role="announce"`
- D) `role="notice"`

Q3. What should the `alt` attribute contain for a decorative image?

- A) The filename
- B) "decorative"
- C) An empty string `alt=""` ✓
- D) The word "image"

Q4. What is the purpose of `tabindex="-1"`?

- A) Makes element the first in tab order
- B) Removes element from tab navigation but allows programmatic focus ✓
- C) Reverses tab order
- D) Makes element non-interactive

Q5. What does `aria-label` do?

- A) Adds a visible label below an element
- B) Provides an accessible name for an element not visible on screen ✓
- C) Labels the entire page
- D) Replaces the `alt` attribute

MODULE 10: Search Engine Optimization (SEO) & Meta Tags

Lesson 1: SEO Head Meta Tags & Social Share Cards

Duration: 25 mins

10.1 What is SEO?

SEO (**Search Engine Optimization**) is the practice of optimizing web pages so they rank higher in search engine results (Google, Bing, etc.).

HTML's role in SEO:

- Page title and meta description
 - Heading structure (H1, H2, H3)
 - Image alt text
 - Semantic HTML structure
 - Page load speed
 - Mobile responsiveness
-

10.2 Essential SEO Meta Tags

```
html

<head>
  <!-- Page title (most important SEO element) -->
  <title>HTML Mastery Course – Learn HTML5 from Scratch | PiyushDhara</title>

  <!-- Meta description (shown in search results) -->
  <meta name="description" content="Complete HTML5 course covering semantic web

  <!-- Keywords (less important today, but still used) -->
  <meta name="keywords" content="HTML course, HTML5, web development, beginner

  <!-- Author -->
  <meta name="author" content="Pankaj Baduwal" />

  <!-- Robots (control indexing) -->
  <meta name="robots" content="index, follow" />

  <!-- Canonical URL (prevents duplicate content) -->
  <link rel="canonical" href="https://www.example.com/html-course" />

  <!-- Charset and viewport (always include) -->
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
</head>
```

10.3 Open Graph Tags (Social Media Cards)

Open Graph tags control how your page appears when shared on Facebook, LinkedIn, WhatsApp:

```
html

<head>
  <!-- Open Graph -->
  <meta property="og:title" content="HTML Mastery Course" />
  <meta property="og:description" content="Learn HTML5 from beginner to advance
  <meta property="og:image" content="https://www.example.com/images/course-thum
  <meta property="og:url" content="https://www.example.com/html-course" />
  <meta property="og:type" content="website" />
  <meta property="og:site_name" content="PiyushDhara" />
  <meta property="og:locale" content="en_US" />
</head>
```

10.4 Twitter Card Tags

```
html

<!-- Twitter Card -->
<meta name="twitter:card" content="summary_large_image" />
<meta name="twitter:site" content="@piyushdhara" />
<meta name="twitter:title" content="HTML Mastery Course" />
<meta name="twitter:description" content="Complete 15-module HTML5 certification course with 15 modules" />
<meta name="twitter:image" content="https://www.example.com/images/twitter-card.jpg" />
```

10.5 Structured Data — Schema.org

```
html

<!-- JSON-LD schema for a course -->
<script type="application/ld+json">
{
  "@context": "https://schema.org",
  "@type": "Course",
  "name": "HTML Mastery: Beginner to Advanced",
  "description": "Complete HTML5 certification course with 15 modules",
  "provider": {
    "@type": "Organization",
    "name": "PiyushDhara"
  },
  "instructor": {
    "@type": "Person",
    "name": "Pankaj Baduwal"
  }
}
</script>
```

10.6 Favicon

```
html
```

```
<!-- Basic favicon -->
<link rel="icon" href="/favicon.ico" />

<!-- PNG favicon (multiple sizes) -->
<link rel="icon" type="image/png" sizes="32x32" href="/favicon-32x32.png" />
<link rel="icon" type="image/png" sizes="16x16" href="/favicon-16x16.png" />

<!-- Apple touch icon -->
<link rel="apple-touch-icon" sizes="180x180" href="/apple-touch-icon.png" />

<!-- Web app manifest -->
<link rel="manifest" href="/site.webmanifest" />
```

10.7 robots.txt

The `robots.txt` file (at the root of the domain) tells search bots what to crawl:

```
# Allow all bots to crawl everything
User-agent: *
Allow: /

# Block a specific folder from indexing
Disallow: /private/

# Block a specific bot
User-agent: Googlebot
Disallow: /temp/

# Link to sitemap
Sitemap: https://www.example.com/sitemap.xml
```

QUIZ — Module 10 (5 Questions)

Q1. Which HTML element has the greatest impact on SEO page ranking?

- A) `<meta name="keywords">`
- B) `<title>` ☒
- C) `<meta name="author">`
- D) `<link rel="canonical">`

Q2. What is the purpose of Open Graph tags?

- A) To improve Google search ranking
- B) To control how pages appear when shared on social media ☒
- C) To add structured data for search engines

- D) To track website analytics

Q3. What does a `<link rel="canonical">` tag do?

- A) Links to the CSS stylesheet
- B) Tells search engines the preferred URL for duplicate content 
- C) Sets the page language
- D) Connects to a CDN

Q4. Which meta robots value prevents a page from being indexed?

- A) `content="no-robots"`
- B) `content="private"`
- C) `content="noindex, nofollow"` 
- D) `content="block"`

Q5. What file tells search engine bots which pages to crawl?

- A) `sitemap.xml`
- B) `seo.txt`
- C) `robots.txt` 
- D) `crawler.conf`

MODULE 11: Responsive Web Design & Viewport Optimization

Lesson 1: Responsive Images & Mobile Optimization

Duration: 20 mins

11.1 The Viewport Meta Tag

html

```
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
```

- `width=device-width` — Sets page width to device screen width
 - `initial-scale=1.0` — Sets initial zoom level to 100%
 - Without this tag, mobile browsers render the desktop version and zoom out
-

11.2 Responsive Images with `srcset`

`srcset` lets the browser choose the best image for the screen resolution:

html

```
<!-- Resolution switching: same image, different sizes -->

```

How it works:

- `srcset` lists image files with their widths (`(w)` descriptor)
- `sizes` tells browser how wide the image will be displayed
- Browser picks the most appropriate image automatically

11.3 Art Direction with `<picture>`

Use `<picture>` when you want completely different images for different screen sizes (different crop, different content):

```
html

<picture>
  <!-- Mobile: portrait crop -->
  <source
    media="(max-width: 480px)"
    srcset="hero-mobile.jpg"
  />

  <!-- Tablet: medium crop -->
  <source
    media="(max-width: 1024px)"
    srcset="hero-tablet.jpg"
  />

  <!-- Desktop: full wide image (fallback img required) -->
  
</picture>
```

11.4 Modern Image Formats with `<picture>`

```
html

<picture>
  <!-- WebP for modern browsers -->
  <source type="image/webp" srcset="photo.webp" />

  <!-- AVIF for cutting-edge browsers -->
  <source type="image/avif" srcset="photo.avif" />

  <!-- JPEG fallback for older browsers -->
  
</picture>
```

11.5 Lazy Loading

```
html

<!-- Load image only when it enters the viewport -->


<!-- Eager (default) – load immediately -->


<!-- Lazy load iframes -->
<iframe src="https://www.youtube.com/embed/..." loading="lazy"></iframe>
```

When to use:

- `loading="lazy"` — Images below the fold (not visible on first load)
- `loading="eager"` — Hero images visible on page load (do NOT lazy load these)

11.6 Flexible Sizing

```
html
```

```
<!-- Always fill container width (responsive) -->


<!-- Responsive video container (16:9 aspect ratio) -->
<div style="position: relative; padding-bottom: 56.25%; height: 0; overflow: hi
  <iframe
    style="position: absolute; top: 0; left: 0; width: 100%; height: 100%;"
    src="https://www.youtube.com/embed/VIDEO_ID"
    allowfullscreen>
  </iframe>
</div>
```

QUIZ — Module 11 (5 Questions)

Q1. What does `width=device-width` in the viewport meta tag do?

- A) Sets the image to device width
- B) Sets the page width to match the device screen width ✓
- C) Locks the page to desktop width
- D) Enables zoom

Q2. When should you use `loading="lazy"` on an image?

- A) On all images
- B) On the hero/banner image at the top
- C) On images below the fold that don't need to load immediately ✓
- D) Never — it's not a real attribute

Q3. What is the purpose of the `srcset` attribute?

- A) Sets multiple source files; browser picks the best for screen size/resolution ✓
- B) Lists backup images if the main one fails
- C) Loads all listed images at once
- D) Sets the image format

Q4. When should you use `<picture>` instead of just `srcset`?

- A) When you want a larger image
- B) When you need art direction — different image content for different screen sizes ✓
- C) When using WebP format
- D) Only for videos

Q5. Which image format provides the best compression while maintaining quality in modern browsers?

- A) JPEG
- B) PNG
- C) GIF
- D) WebP 

MODULE 12: Advanced HTML5 Web APIs & Local Storage

Lesson 1: HTML5 Canvas, Web Storage & Geolocation

Duration: 30 mins

12.1 HTML5 Canvas

The `<canvas>` element creates a drawing surface controlled by JavaScript:

html

```
<canvas id="myCanvas" width="500" height="300"></canvas>

<script>
  const canvas = document.getElementById('myCanvas');
  const ctx = canvas.getContext('2d');

  // Draw a filled rectangle
  ctx.fillStyle = '#3498db';
  ctx.fillRect(50, 50, 200, 100);

  // Draw a circle
  ctx.beginPath();
  ctx.arc(300, 100, 50, 0, 2 * Math.PI);
  ctx.fillStyle = '#e74c3c';
  ctx.fill();

  // Draw text
  ctx.font = '24px Arial';
  ctx.fillStyle = '#2c3e50';
  ctx.fillText('Hello Canvas!', 50, 250);

  // Draw a line
  ctx.beginPath();
  ctx.moveTo(0, 150);
  ctx.lineTo(500, 150);
  ctx.strokeStyle = '#27ae60';
  ctx.lineWidth = 2;
  ctx.stroke();
</script>
```

Canvas Use Cases:

- Data visualizations and charts
- Games and animations
- Image editing tools
- Drawing applications

12.2 Inline SVG

SVG (Scalable Vector Graphics) is written directly in HTML:

```
html

<svg width="200" height="200" xmlns="http://www.w3.org/2000/svg">

  <!-- Circle -->
  <circle cx="100" cy="100" r="80" fill="#3498db" stroke="#2980b9" stroke-width

  <!-- Rectangle -->
  <rect x="20" y="20" width="60" height="40" fill="#e74c3c" rx="5" />

  <!-- Text -->
  <text x="100" y="105" text-anchor="middle" fill="white" font-size="20">SVG</t

  <!-- Line -->
  <line x1="10" y1="10" x2="190" y2="190" stroke="#27ae60" stroke-width="2" />

  <!-- Polygon (triangle) -->
  <polygon points="100,10 190,190 10,190" fill="#f39c12" />

</svg>
```

SVG vs Canvas:

SVG	Canvas
Vector (scalable)	Raster (pixel-based)
DOM elements	Drawn on a bitmap
Good for icons, logos	Good for games, animations
Events on elements	Events on whole canvas

12.3 Web Storage API

The Web Storage API stores data in the browser. Unlike cookies, data is not sent to the server.

localStorage — Persists until manually cleared:

```
javascript

// Store data
localStorage.setItem('username', 'Pankaj');
localStorage.setItem('theme', 'dark');

// Retrieve data
const name = localStorage.getItem('username'); // "Pankaj"
const theme = localStorage.getItem('theme');    // "dark"

// Store an object (must convert to JSON string)
const user = { name: 'Pankaj', level: 'Advanced' };
localStorage.setItem('user', JSON.stringify(user));
const savedUser = JSON.parse(localStorage.getItem('user'));

// Remove a specific item
localStorage.removeItem('theme');

// Clear all localStorage
localStorage.clear();
```

sessionStorage — Cleared when tab/browser closes:

```
javascript

// Same API as localStorage
sessionStorage.setItem('cart', JSON.stringify(cartItems));
const cart = JSON.parse(sessionStorage.getItem('cart'));
sessionStorage.removeItem('cart');
```

Comparison:

Feature	localStorage	sessionStorage	Cookies
Capacity	~5MB	~5MB	~4KB
Persists	Until cleared	Until tab closes	Set expiry
Sent to server	No	No	Yes
Accessible from JS	Yes	Yes	Yes

12.4 Geolocation API

javascript

```
// Check if browser supports Geolocation
if ('geolocation' in navigator) {

    // Get current position (asks user for permission)
    navigator.geolocation.getCurrentPosition(
        // Success callback
        function(position) {
            const lat = position.coords.latitude;
            const lng = position.coords.longitude;
            const accuracy = position.coords.accuracy;

            console.log(`Latitude: ${lat}`);
            console.log(`Longitude: ${lng}`);
            console.log(`Accuracy: ${accuracy} meters`);
        },
        // Error callback
        function(error) {
            switch(error.code) {
                case error.PERMISSION_DENIED:
                    console.log('User denied location access');
                    break;
                case error.POSITION_UNAVAILABLE:
                    console.log('Location unavailable');
                    break;
                case error.TIMEOUT:
                    console.log('Request timed out');
                    break;
            }
        },
        // Options
        {
            enableHighAccuracy: true,
            timeout: 5000,
            maximumAge: 0
        }
    );

    // Watch position (updates continuously)
    const watchId = navigator.geolocation.watchPosition(successCallback, errorCallback);

    // Stop watching
    navigator.geolocation.clearWatch(watchId);

} else {
```

```
console.log('Geolocation is not supported by this browser');
}
```

12.5 Other HTML5 APIs (Overview)

Drag and Drop API:

```
html

<div draggable="true" ondragstart="drag(event)" id="item">Drag me</div>
<div ondrop="drop(event)" ondragover="allowDrop(event)">Drop here</div>
```

History API:

```
javascript

// Add entry to browser history
history.pushState({ page: 1 }, "Title", "/page1");

// Replace current entry
history.replaceState({ page: 2 }, "Title", "/page2");

// Navigate
history.back();
history.forward();
```

Notification API:

```
javascript

Notification.requestPermission().then(permission => {
  if (permission === 'granted') {
    new Notification('Hello!', {
      body: 'You have a new message.',
      icon: '/icon.png'
    });
  }
});
```

QUIZ — Module 12 (5 Questions)

Q1. What method is used to get the 2D drawing context from a canvas element?

- A) `canvas.getContext('2d')` ☒
- B) `canvas.draw2D()`
- C) `canvas.context('2d')`
- D) `canvas.get2DContext()`

Q2. What is the main difference between `localStorage` and `sessionStorage`?

- A) `localStorage` stores more data
- B) `sessionStorage` is more secure
- C) `localStorage` persists until cleared; `sessionStorage` clears when the tab closes ✓
- D) They are identical

Q3. How must objects be stored in `localStorage`?

- A) As-is using direct assignment
- B) Converted to a JSON string using `JSON.stringify()` ✓
- C) Using the `.store()` method
- D) As binary data

Q4. Which API asks the user's permission to access device location?

- A) `navigator.location`
- B) `window.position`
- C) `navigator.geolocation` ✓
- D) `document.geo`

Q5. Which is better for scalable logos and icons — Canvas or SVG?

- A) Canvas, because it's pixel-perfect
- B) SVG, because it's vector-based and scales without quality loss ✓
- C) Canvas, because it's faster
- D) They are equivalent

MODULE 13: Web Standards, Validation & Clean Code

Lesson 1: W3C Code Validation & Best Practices

Duration: 20 mins

13.1 What is W3C Validation?

The **W3C (World Wide Web Consortium)** is the international body that sets web standards. The W3C Markup Validator checks your HTML for errors against these standards.

Validate at: <https://validator.w3.org>

Why validate?

- Catches typos and missing closing tags
 - Ensures cross-browser compatibility
 - Improves accessibility
 - Required for professional development
-

13.2 Common HTML Validation Errors

html

```
<!-- ✖ Error: Missing required alt attribute -->

```

```
<!-- ✔ Fixed -->

```

```
<!-- ✖ Error: Element not closed properly -->
<p>This paragraph is not closed
<p>This starts a new paragraph
```

```
<!-- ✔ Fixed -->
<p>This paragraph is closed correctly.</p>
<p>This starts a new paragraph.</p>
```

```
<!-- ✖ Error: Invalid nesting (block inside inline) -->
<a href="#"><div>Click me</div></a>
```

```
<!-- ✔ Fixed -->
<div><a href="#">Click me</a></div>
```

```
<!-- ✖ Error: Duplicate ID -->
<div id="header">...</div>
<div id="header">...</div>
```

```
<!-- ✔ Fixed: IDs must be unique -->
<div id="main-header">...</div>
<div id="sub-header">...</div>
```

13.3 HTML Best Practices

1. Always declare DOCTYPE:

html

```
<!DOCTYPE html>
```

2. Use lowercase for tags and attributes:

html


```
<!-- ❌ Old style -->
<P CLASS="intro">Hello</P>

<!-- ✅ Modern style -->
<p class="intro">Hello</p>
```

3. Always quote attribute values:

```
html

<!-- ❌ Avoid -->
<img src=photo.jpg alt=Photo />

<!-- ✅ Always quote -->

```

4. Proper indentation (2 or 4 spaces):

```
html

<body>
  <header>
    <nav>
      <ul>
        <li><a href="/">Home</a></li>
      </ul>
    </nav>
  </header>
  <main>
    <h1>Page Title</h1>
  </main>
</body>
```

5. Close all tags:

```
html

<!-- ✅ Always close tags -->
<p>Paragraph content.</p>
<li>List item</li>
```

6. Use semantic HTML over generic divs:

```
html
```

```
<!-- ❌ Non-semantic -->
<div class="header"><div class="nav">...</div></div>

<!-- ✅ Semantic -->
<header><nav>...</nav></header>
```

7. Add `lang` to `<html>` tag:

```
html

<html lang="en">
```

8. One H1 per page:

```
html

<h1>One main page title</h1>
<h2>Section title</h2>
<h3>Sub-section</h3>
```

13.4 Cross-Browser Testing

Common tools:

- **BrowserStack** — Test on real browsers online
- **Can I Use** (caniuse.com) — Check browser support for HTML/CSS features
- **Chrome DevTools** — Device emulation for mobile testing
- **Firefox Developer Tools** — Grid and flexbox inspectors

Testing checklist:

- Chrome (latest)
- Firefox (latest)
- Safari (latest)
- Edge (latest)
- Mobile Chrome (Android)
- Mobile Safari (iOS)

13.5 HTML Code Comments and Organization

```
html
```

```
<!-- ===== HEADER SECTION ===== -->
<header>
  <nav>...</nav>
</header>

<!-- ===== MAIN CONTENT ===== -->
<main>

  <!-- Hero Section -->
  <section id="hero">
    <h1>Welcome</h1>
  </section>

  <!-- Features Section -->
  <section id="features">
    <!-- TODO: Add feature cards here -->
  </section>

</main>

<!-- ===== FOOTER ===== -->
<footer>...</footer>
```

QUIZ — Module 13 (5 Questions)

Q1. Which organization maintains web standards and provides the HTML validator?

- A) IEEE
- B) ISO
- C) W3C ☒
- D) IETF

Q2. What will the W3C validator report if you use the same `id` twice on a page?

- A) A warning only
- B) Nothing — it's allowed
- C) An error — IDs must be unique ☒
- D) It auto-corrects to class names

Q3. What is the correct practice for attribute values in HTML5?

- A) Always use single quotes
- B) Quotes are optional for all attributes
- C) Always use double quotes ☒
- D) Never use quotes

Q4. What tool shows browser support for HTML5 features?

- A) W3C Validator
- B) Can I Use (caniuse.com) ✓
- C) Lighthouse
- D) Web.dev

Q5. What is the recommended indentation for nested HTML elements?

- A) Tabs only
- B) No indentation needed
- C) 2 or 4 spaces consistently ✓
- D) 8 spaces

MODULE 14: Real-World Portfolio & Capstone Projects

Lesson 1: Building a Complete Personal Portfolio Webpage

Duration: 45 mins

14.1 Project Overview

In this module, you will build **three real-world projects** using everything you have learned:

1. **Personal Portfolio Website** — Multi-section single-page site
 2. **Restaurant Website** — Multi-page business site
 3. **Personal Resume** — Formatted HTML resume
-

14.2 Project 1: Personal Portfolio Website

File structure:

```
portfolio/  
├─ index.html  
├─ about.html  
├─ projects.html  
├─ contact.html  
├─ css/  
│   └─ style.css  
└─ images/  
    ├── profile.jpg  
    └─ project-1.png
```

Complete `index.html`:

```
html
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Pankaj Baduwal – Frontend Developer and HTML Educator" />
  <meta property="og:title" content="Pankaj Baduwal | Frontend Developer" />
  <meta property="og:image" content="images/profile.jpg" />
  <title>Pankaj Baduwal | Frontend Developer</title>
  <link rel="stylesheet" href="css/style.css" />
</head>
<body>

  <!-- Skip Navigation -->
  <a href="#main" class="skip-link">Skip to main content</a>

  <!-- Header & Navigation -->
  <header role="banner">
    <div class="logo">
      <a href="index.html">PB<span>.</span></a>
    </div>
    <nav role="navigation" aria-label="Main navigation">
      <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="about.html">About</a></li>
        <li><a href="projects.html">Projects</a></li>
        <li><a href="contact.html">Contact</a></li>
      </ul>
    </nav>
  </header>

  <!-- Main Content -->
  <main id="main" role="main">

    <!-- Hero Section -->
    <section id="hero" aria-labelledby="hero-title">
      <figure>
        
      </figure>
      <div class="hero-text">
        <h1 id="hero-title">Hi, I'm <strong>Pankaj Baduwal</strong></h1>
        <p>Frontend Developer & HTML Educator based in Kathmandu, Nepal.</p>
        <a href="projects.html" class="btn">View My Work</a>
        <a href="contact.html" class="btn btn-outline">Contact Me</a>
      </div>
    </section>

  </main>

</body>
</html>
```

```
<!-- Skills Section -->
<section id="skills" aria-labelledby="skills-title">
  <h2 id="skills-title">My Skills</h2>
  <ul class="skills-list">
    <li>HTML5</li>
    <li>CSS3</li>
    <li>JavaScript</li>
    <li>Responsive Design</li>
    <li>Web Accessibility</li>
    <li>SEO</li>
  </ul>
</section>
```

```
<!-- Featured Projects -->
<section id="projects" aria-labelledby="projects-title">
  <h2 id="projects-title">Featured Projects</h2>
  <div class="project-grid">
    <article class="project-card">
      <figure>
        
        <figcaption>Himalayan Bites Restaurant Website</figcaption>
      </figure>
      <h3>Himalayan Bites</h3>
      <p>A responsive multi-page restaurant website built with semantic HTML5</p>
      <a href="projects.html#restaurant" aria-label="View Himalayan Bites project" />View Project</a>
    </article>
  </div>
</section>
```

```
</main>
```

```
<!-- Footer -->
<footer role="contentinfo">
  <address>
    <p>Contact: <a href="mailto:pankaj@example.com">pankaj@example.com</a></p>
  </address>
  <p>&copy; <time datetime="2024">2024</time> Pankaj Baduwal. All rights reserved.</p>
  <nav aria-label="Social media links">
    <ul>
      <li><a href="https://github.com/pankaj" target="_blank" rel="noopener noreferrer">GitHub</a></li>
      <li><a href="https://linkedin.com/in/pankaj" target="_blank" rel="noopener noreferrer">LinkedIn</a></li>
    </ul>
  </nav>
</footer>
```

```
</body>
</html>
```

14.3 Project 2: Restaurant Website (Key Pages)

`menu.html` excerpt:

```
html

<main>
  <h1>Our Menu</h1>

  <section aria-labelledby="starters-title">
    <h2 id="starters-title">Starters</h2>
    <dl>
      <dt>Momo (8 pcs) – <strong>NPR 250</strong></dt>
      <dd>Steamed or fried dumplings filled with chicken or vegetables, served

      <dt>Sekewa – <strong>NPR 350</strong></dt>
      <dd>Grilled marinated meat on skewers with spicy dipping sauce.</dd>
    </dl>
  </section>

  <section aria-labelledby="mains-title">
    <h2 id="mains-title">Main Course</h2>
    <!-- More menu items... -->
  </section>
</main>
```

14.4 Project 3: HTML Resume

```
html
```

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>Pankaj Baduwal – Resume</title>
</head>
<body>

  <header>
    <h1>Pankaj Baduwal</h1>
    <address>
      <a href="mailto:pankaj@example.com">pankaj@example.com</a> |
      <a href="tel:+9779800000000">+977-980-000-0000</a> |
      Kathmandu, Nepal
    </address>
  </header>

  <main>
    <section>
      <h2>Professional Summary</h2>
      <p>Passionate frontend developer and educator with 5+ years of experience</p>
    </section>

    <section>
      <h2>Skills</h2>
      <ul>
        <li>HTML5, CSS3, JavaScript</li>
        <li>Responsive Design, Web Accessibility (WCAG 2.1)</li>
        <li>SEO, Web Performance Optimization</li>
      </ul>
    </section>

    <section>
      <h2>Experience</h2>
      <article>
        <h3>Lead Educator – PiyushDhara</h3>
        <p><time datetime="2022-01">January 2022</time> – Present</p>
        <ul>
          <li>Designed and delivered the HTML Mastery 15-module certification c</li>
          <li>Mentored 500+ students in web development</li>
        </ul>
      </article>
    </section>

    <section>
      <h2>Education</h2>
```



```
<dl>
  <dt>Bachelor of Computer Science</dt>
  <dd>Tribhuvan University, Kathmandu - <time datetime="2019">2019</time>
</dl>
</section>
</main>

</body>
</html>
```

QUIZ — Module 14 (5 Questions)

Q1. In a portfolio project structure, where should CSS files be placed?

- A) In the root folder next to HTML files
- B) In a dedicated `css/` subfolder ☒
- C) Inside the `<head>` tag as inline styles only
- D) In the `images/` folder

Q2. Why should you use `<article>` for each project card in a portfolio?

- A) It makes cards look better
- B) Each project is self-contained content that makes sense independently ☒
- C) `<article>` adds built-in styling
- D) `<div>` tags don't work for cards

Q3. Which element should be used for contact information in a resume footer?

- A) `<contact>`
- B) `<p>`
- C) `<address>` ☒
- D) `<footer>`

Q4. What attribute should be added to images in a project gallery below the fold?

- A) `loading="eager"`
- B) `loading="lazy"` ☒
- C) `async="true"`
- D) `defer="true"`

Q5. What is the purpose of `aria-label="Pankaj's GitHub profile"` on a social media link?

- A) It adds a tooltip that appears on hover
- B) It provides descriptive text for screen reader users who can't see the icon/text ☒
- C) It improves the link's Google ranking
- D) It opens the link in a new tab

MODULE 15: Final Assessment & Professional Certification Exam

Lesson 1: Certification Exam Preparation & Summary

Duration: 30 mins

15.1 Course Summary

Congratulations on completing all 14 modules! Here is a complete recap of everything you have learned:

Foundation (Modules 1-3):

- How the internet and client-server architecture works
- HTML document structure, DOCTYPE, boilerplate
- All text formatting elements and when to use them
- Creating links: absolute, relative, email, phone, bookmark

Content Elements (Modules 4-6):

- Images with proper alt text, audio, video, iframes
- Unordered, ordered, and description lists with nesting
- Complex data tables with rowspan, colspan, and accessibility

Interactive Features (Module 7):

- All form input types and their HTML5 validation
- Labels, fieldsets, and accessible form structure

Advanced Structure (Modules 8-9):

- Full semantic HTML5 page architecture
- ARIA roles, states, and properties
- WCAG 2.1 accessibility principles

Optimization (Modules 10-11):

- SEO meta tags, Open Graph, Twitter Cards
- Responsive images with srcset and `<picture>`
- Lazy loading and viewport optimization

Advanced APIs (Module 12):

- HTML5 Canvas drawing API
- SVG inline graphics
- localStorage and sessionStorage
- Geolocation API

Professional Skills (Modules 13-14):

- W3C validation and code best practices
 - Cross-browser testing
 - Three complete real-world projects
-

15.2 Key Formulas to Remember

```
html

<!-- Perfect HTML5 Boilerplate -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <meta name="description" content="Page description for SEO" />
  <title>Page Title</title>
</head>
<body>
  <header>
    <nav>...</nav>
  </header>
  <main>
    <article>...</article>
    <aside>...</aside>
  </main>
  <footer>...</footer>
</body>
</html>
```

15.3 Self-Assessment Checklist

Before the exam, verify you can do all of the following from memory:

HTML Structure:

- ☐ Write a complete HTML5 boilerplate
- ☐ Explain every tag in the `<head>` section
- ☐ Structure a page with semantic HTML elements

Text & Links:

- ☐ Use all six heading levels correctly
- ☐ Create bold, italic, subscript, superscript text
- ☐ Create links that open in new tabs safely

Media:

- ☐ Embed a responsive image with alt text

- ☐ Add an HTML5 video with controls and poster
- ☐ Embed a YouTube video with iframe

Forms:

- ☐ Create a form with at least 5 different input types
- ☐ Add proper labels and fieldsets
- ☐ Use HTML5 validation attributes

Accessibility:

- ☐ Add ARIA roles to landmark elements
- ☐ Use aria-label on icon buttons
- ☐ Create a skip navigation link

SEO:

- ☐ Write all essential meta tags
- ☐ Add Open Graph tags for social sharing

FINAL CERTIFICATION EXAM

30 Questions — Covering All 15 Modules

Q1. What is the first line of every HTML5 document?

- A) `<html>`
- B) `<head>`
- C) `<!DOCTYPE html>` 
- D) `<!-- HTML5 -->`

Q2. Which tag creates the largest heading on a page?

- A) `<h6>`
- B) `<heading>`
- C) `<h1>` 
- D) `<big>`

Q3. What attribute must every `<img>` tag include for accessibility?

- A) `title`
- B) `src`
- C) `alt` 
- D) `width`

Q4. What does the `<strong>` tag convey that `<b>` does not?

- A) Stronger visual boldness

- B) Semantic importance for screen readers ☒
- C) Bold text in all browsers
- D) They are identical

Q5. Which HTML element is used to group radio buttons with a label?

- A) `<group>`
- B) `<section>`
- C) `<fieldset>` ☒
- D) `<form>`

Q6. What is the correct way to create a link that opens an email client?

- A) `<a href="email:user@example.com">`
- B) `<a href="mail:user@example.com">`
- C) `<a href="mailto:user@example.com">` ☒
- D) `<a href="send:user@example.com">`

Q7. Which attribute in `<ol>` makes the list count in reverse?

- A) `order="reverse"`
- B) `reversed` ☒
- C) `type="reverse"`
- D) `direction="rtl"`

Q8. How do you make a table cell span 3 columns?

- A) `<td span="3">`
- B) `<td columns="3">`
- C) `<td colspan="3">` ☒
- D) `<td width="3">`

Q9. Which semantic element should contain the main navigation of a website?

- A) `<menu>`
- B) `<ul>`
- C) `<navigation>`
- D) `<nav>` ☒

Q10. What does `role="alert"` do in ARIA?

- A) Styles the element red
- B) Announces content changes to screen readers immediately ☒
- C) Adds a browser alert popup
- D) Marks content as important

Q11. Which meta tag is most critical for mobile responsiveness?

- A) `<meta name="mobile" content="true">`
- B) `<meta name="viewport" content="width=device-width, initial-scale=1.0">` ✓
- C) `<meta name="screen" content="responsive">`
- D) `<meta name="device" content="mobile">`

Q12. Which Open Graph property sets the image that appears when sharing on social media?

- A) `og:thumbnail`
- B) `og:photo`
- C) `og:image` ✓
- D) `og:picture`

Q13. What does `<abbr title="World Wide Web">WWW</abbr>` do?

- A) Creates a link to a definition
- B) Shows the full form as a tooltip on hover ✓
- C) Makes the text bold
- D) Adds a footnote

Q14. Which input type shows a color picker in the browser?

- A) `type="picker"`
- B) `type="hue"`
- C) `type="color"` ✓
- D) `type="rgb"`

Q15. What is the purpose of `<figcaption>`?

- A) To add a border to images
- B) To provide a visible caption for a `<figure>` element ✓
- C) To hide image alt text
- D) To link images to other pages

Q16. Where does localStorage data persist?

- A) On the web server
- B) In a cookie with expiry
- C) In the browser until manually cleared ✓
- D) Only for the current browser session

Q17. What does `srcset` do on an `<img>` tag?

- A) Provides multiple source files; the browser selects the best for screen size ✓
- B) Adds a backup image if the main one fails
- C) Sets the image source for IE browsers

- D) Loads all listed images simultaneously

Q18. Which W3C tool checks your HTML for errors and standards compliance?

- A) Chrome DevTools
- B) W3C Markup Validator ☒
- C) Can I Use
- D) Lighthouse

Q19. What tag is used to display a horizontal dividing line?

- A) `<line>`
- B) `<divider>`
- C) `<br />`
- D) `<hr />` ☒

Q20. Which attribute links a `<label>` to an `<input>`?

- A) `name` on both elements
- B) `for` on the label matching `id` on the input ☒
- C) `link` on the label
- D) `target` on the input

Q21. What does `loading="lazy"` do on an image?

- A) Makes the image load in low quality first
- B) Defers image loading until it's near the viewport ☒
- C) Loads the image in a background thread
- D) Lazy loads the entire page

Q22. Which Canvas method begins a new path for drawing?

- A) `ctx.newPath()`
- B) `ctx.startPath()`
- C) `ctx.beginPath()` ☒
- D) `ctx.createPath()`

Q23. What does `aria-expanded="false"` communicate?

- A) The element has no content
- B) A collapsible region is currently closed ☒
- C) The element is disabled
- D) The element is hidden

Q24. In which section of a table should summary or total rows appear?

- A) `<thead>`
- B) `<tbody>`

- C) `<tfoot>` ☒
- D) `<tsummary>`

Q25. What is the purpose of `<link rel="canonical">`?

- A) Links to the main CSS file
- B) Tells search engines the preferred URL to avoid duplicate content penalties ☒
- C) Connects the page to a CDN
- D) Marks the page as the main page of the site

Q26. What does `tabindex="-1"` do to an element?

- A) Makes it the first element in tab order
- B) Removes it from keyboard tab navigation but allows programmatic focus ☒
- C) Makes it non-interactive permanently
- D) Adds it to a reversed tab order

Q27. Which `<ol>` attribute controls where numbering starts?

- A) `begin`
- B) `from`
- C) `start` ☒
- D) `number`

Q28. Which of these is NOT a valid HTML5 semantic element?

- A) `<article>`
- B) `<section>`
- C) `<content>` ☒
- D) `<aside>`

Q29. What happens to data in `sessionStorage` when the browser tab is closed?

- A) It is saved to `localStorage` automatically
- B) It remains until the browser is closed
- C) It is cleared immediately ☒
- D) It is sent to the server

Q30. Which Geolocation API method continuously tracks position changes?

- A) `navigator.geolocation.getCurrentPosition()`
 - B) `navigator.geolocation.trackPosition()`
 - C) `navigator.geolocation.watchPosition()` ☒
 - D) `navigator.geolocation.followPosition()`
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FINAL EXAM ANSWER KEY

Q	Answer	Module
1	C	M1
2	C	M2
3	C	M4
4	B	M2
5	C	M7
6	C	M3
7	B	M5
8	C	M6
9	D	M8
10	B	M9
11	B	M11
12	C	M10
13	B	M2
14	C	M7
15	B	M4
16	C	M12
17	A	M11
18	B	M13
19	D	M2
20	B	M7
21	B	M11
22	C	M12
23	B	M9
24	C	M6

Q	Answer	Module
25	B	M10
26	B	M9
27	C	M5
28	C	M8
29	C	M12
30	C	M12

End of HTML Mastery: Beginner to Advanced — Complete Course Content Total: 15 Modules / 75 Module Quiz Questions / 30 Final Exam Questions Prepared for PiyushDhara by Pankaj Baduwal